7-1 Computer Configuration

Point!

The Five Major Components of a Computer

(1) ([1]**Hardware**) : Devices such as the main computer unit and peripheral devices.

(2) Peripheral devices : Devices used by connecting them to a computer.
<Examples> Keyboard, mouse, display, printer, etc.

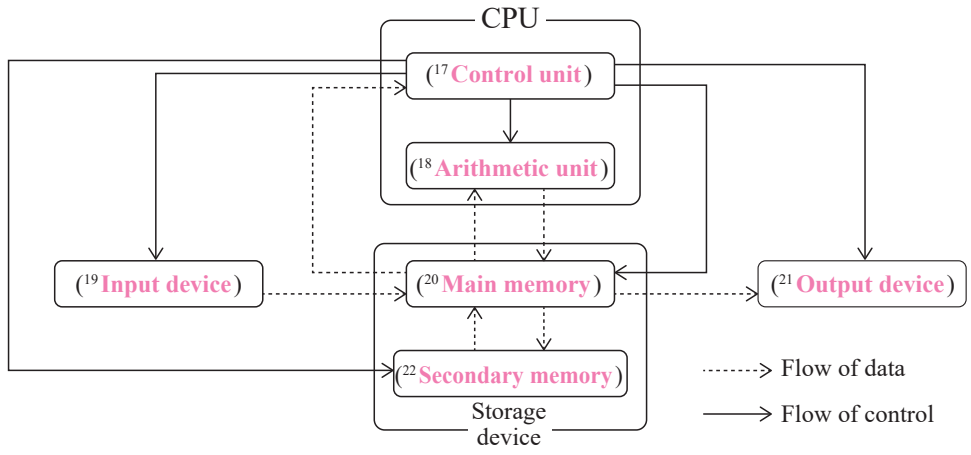
(3) The five major components of a computer : A computer is comprised of the following five components: ([2]**the control unit**), ([3]**the arithmetic unit**), ([4]**the memory unit**), ([5]**the input unit**), and ([6]**the output unit**).

- The control unit and arithmetic unit together are referred to as the ([7]**CPU (central processing unit)**).
- Storage devices are divided into ([8]**main memory**), which temporarily stores programs and data, and ([9]**secondary memory**), which is used for long-term storage.

<Roles of the Five Major Components of a Computer>

Five major components		Role	Example of major component
Control unit	CPU	Executes a computer's ([10]**instructions**) and issues ([11]**commands**) to each function.	CPU
Arithmetic unit		Performs ([12]**calculations**).	
Storage device	Main memory device (Memory)	([13]**Temporarily**) stores programs, data, etc.	Main memory
	Secondary memory (Storage)	For ([14]**long-term**) storage of programs and data.	Hard disk, SSD, USB memory
Input device		([15]**Inputs**) information from external sources.	Keyboard, mouse, scanner
Output device		([16]**Outputs**) information outside of the computer.	Display, printer

<Relationship of the Five Major Components of a Computer>



(4) ([23]**Interface**) : A component that mediates the exchange of information. It is used for connecting the main computer and peripheral devices.

[1] ([24]**USB**) : The most widely used interface for computer peripheral devices.
Compatible with printers, keyboards, and external hard drives.

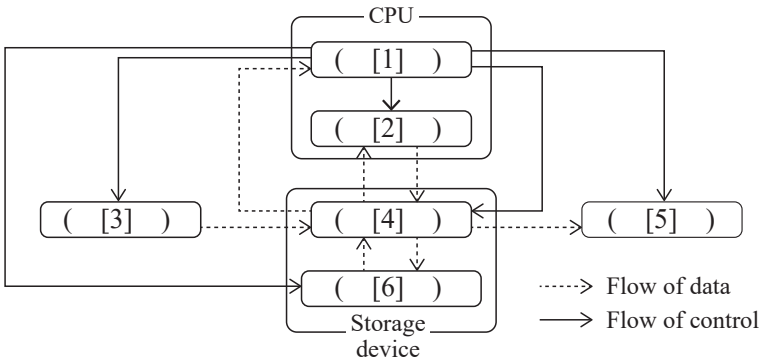
[2] ([25]**HDMI**) : A communication standard that allows video, audio, and other data to be transmitted through a single cable.
Compatible with digital televisions and audio equipment.

[3] ([26]**Ethernet**) : A communication standard used in wired LANs at home, the office, etc.
Compatible with devices such as hubs and routers.

Warm Up

Answer the following questions.

(1) The following diagram represents the relationship between computer components. Choose the appropriate device name that best fits into the blanks [1] to [6] from the options **A** to **F**, and answer using the letters.



A Input device **B** Output device **C** Control unit
D Arithmetic unit **E** Main memory device **F** Secondary memory

(2) Choose all of the following options from **A** to **H** that are output devices, and answer using the letters.

A Memory **B** Display **C** SSD **D** Mouse
E Keyboard **F** Printer **G** Hard disk **H** USB memory

Explanation

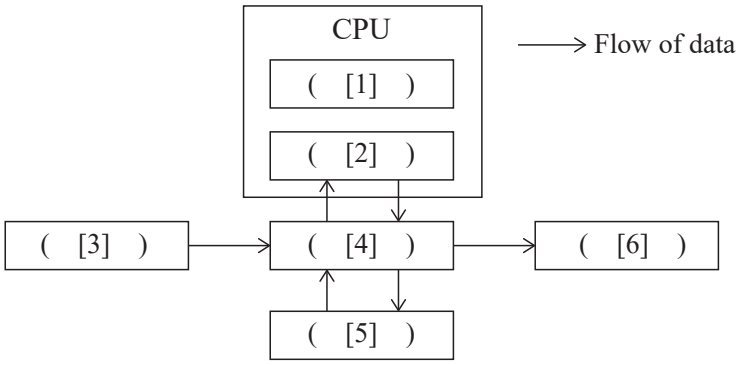
(1) [1] Control unit, because the flow of control is directed towards all devices. **C**
[2] Among the components of the CPU, [1] is the control unit; therefore, the arithmetic unit. **D**
[3] The starting point of the data flow; therefore, the input device. **A**
[4] Among storage devices, this is identified as the main memory because data flows directly into it from the input device. **E**
[5] Marks the end of the data flow; therefore, an output device. **B**
[6] Among storage devices, [4] is the main memory; therefore, the secondary memory. **F**

(2) **B, F**

Try

Answer the following questions.

(1) The following diagram represents the relationship between computer components. Choose the appropriate device name that best fits into the blanks [1] to [6] from the options **A** to **F**, and answer using the letters. Note that the order of blanks [1] and [2] does not matter.



A Input device **B** Output device **C** Control unit
D Arithmetic unit **E** Main memory device **F** Secondary memory

(2) Choose all computer components or peripheral devices corresponding to the following [1] to [5] from the options **A** to **I** below, and answer using the letters.

[1] Arithmetic unit, control unit [2] Main memory device [3] Secondary memory
[4] Input device [5] Output device

A Memory **B** Display **C** SSD
D Mouse **E** Keyboard **F** Printer
G Hard disk **H** USB memory **I** CPU

(3) Choose the term that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below, and answer using the letters.

([1]) is called the main memory and exchanges data directly with the CPU. ([2]) is also called secondary memory and serves the role of storing large amounts of data.

A Memory **B** Hardware
C Interface **D** Storage

(4) What is the interface described by each of the following [1] to [3]?

[1] An interface that connects devices to output video and audio through a single cable.
[2] The most commonly used interface for computer peripheral devices.
[3] Communication standard used for wired LANs in homes, offices, etc.

Exercise

1 Answer the following questions.

(1) Choose the most appropriate explanation of the control unit, which is one of the five major components of a computer from the options **A** to **D** below, and answer using the letter.

A It is a device that performs calculations.
B It is a device that executes a computer's instructions and issues commands to each function.
C It is a device for temporarily storing programs and data.
D It is a device for inputting information from external sources.

(2) Choose the names of the devices that correspond to computer devices and peripheral devices [1] to [5] from the options **A** to **E** below, and answer using the letters.

[1] Display [2] HDD [3] Keyboard
[4] CPU [5] Main memory

A Input device **B** Output device **C** Control unit, arithmetic unit
D Main memory device **E** Secondary memory

(3) Choose all the peripheral devices from the options **A** to **F**, and answer using the letters.

A Display **B** SSD **C** Memory
D CPU **E** Keyboard **F** USB memory

(4) Choose one correct statement regarding interfaces from the options **A** to **D** below, and answer using the letter.

A Devices that are not the computer itself but are used by connecting to a computer.
B Devices such as the computer itself and peripheral devices.
C Devices for temporarily storing programs and data.
D A component that mediates the exchange of information. It is used for connecting the main computer and peripheral devices.

(5) Choose the corresponding interface for each of the following peripheral devices [1] to [3] from the options **A** to **D**, and answer using the letters.

[1] Keyboard [2] Router [3] Audio equipment

A Ethernet **B** HDMI **C** SSD **D** USB